

Usability Audit Report

PHC Queue Management System — Challenge 1.3, Track B
Pilot site: Latha Children’s Clinic, Attapur, Hyderabad

1. Methodology and Scope

This report documents a developer self-test usability audit conducted across four distinct user personas defined in the product requirements: Patient (smartphone-literate), Front Desk Staff, Doctor / Medical Officer, and Cold-Start User. The audit was conducted by the developer playing each role in turn, using a structured interview format with task-based scoring on a five-point scale (1 = unable to complete, 5 = completed instantly with full confidence). A five-user audit with real PHC patients and staff is scheduled for pilot week one and is not in scope for this hackathon submission.

Scope note: V1 ships without automated WhatsApp or SMS notifications — phone numbers are captured at token issue and surfaced on the front-desk dashboard so staff can call back patients who are missing when called. Findings touching automated message clarity inform V2 planning; V1-specific findings drive pre-pilot fixes.

A task is considered ‘positive’ if its score is greater than or equal to 3.5. A persona is considered ‘satisfied’ if at least 75 percent of its tasks are positive. The overall pass criterion is at least 80 percent of personas satisfied.

2. Per-Persona Results

Persona 1: Patient, Smartphone-Literate

Task	Score	Verbatim Note
Read token confirmation WhatsApp	4.5	Clear on token number, wait, leave-and-return, and STATUS keyword. Concern raised about token number.
Reply STATUS to check position	4.0	Trivial to remember. Specifically asked whether the system handles lower-case input.
Interpret 3-ahead alert	5.0	Message conveys clear urgency and action.

Tasks positive: 3 of 3. Persona satisfied: yes.

Persona 2: Front Desk Staff

Task	Score	Verbatim Note
Open OPD session	4.0	Button is clear, but first-time staff with no tech exposure may hesitate without a tooltip.
Issue 5 tokens (3 with phone, 2 without)	5.0	Fast in practice. Phrasing ‘token is the new mobile phone’ from interview.
Identify waiting count	5.0	Visible at a glance from the navy panel.
Notice doctor-inactive alert	4.0	Alert is clear, but the staff member would benefit from seeing elapsed time and an estimated wait time.

Tasks positive: 4 of 4. Persona satisfied: yes.

Persona 3: Doctor / Medical Officer

Task	Score	Verbatim Note
Read current-serving display	4.0	Bit clicky for a clinician’s taste but acceptable within the deployment context.

Tap Next Patient	4.0	Flow is clear but the button currently risks double-fire on inadvertent double-tap. I
Read stats footer	3.0	Average consultation time was called out as low-value to the doctor; remaining co
End-of-day empty state	5.0	Closure is clear and minimal.

Tasks positive: 3 of 4 (75 percent). Persona satisfied: yes.

Persona 4: Cold-Start User

Task	Score	Verbatim Note
Issue a token unaided	4.0	Layout reads correctly; Telugu or Hindi labels would materially improve confidence
Advance queue unaided (doctor screen)	5.0	Single primary action is self-explanatory.
Grandma-test summary	No	A non-tech-literate elderly user would struggle. Smartphone proficiency and English

Tasks positive: 2 of 2 scored tasks. Persona satisfied: yes (grandma-test counted as qualitative finding, not pass/fail task).

3. Overall Satisfaction

Persona	Tasks Positive	Satisfied
P1 Patient	3 of 3 (100%)	Yes
P3 Front Desk	4 of 4 (100%)	Yes
P4 Doctor	3 of 4 (75%)	Yes
P5 Cold Start	2 of 2 (100%)	Yes
TOTAL	4 of 4 personas (100%)	Pass

Pass criterion (at least 80 percent of personas satisfied) is met. Result subject to confirmation in the scheduled five-user pilot audit.

4. Findings and Triage

Six distinct findings were surfaced during the audit. Each is assigned a triage decision: Will Fix (scheduled for v1.1 before pilot), Open (under consideration, no commitment), or Won't Fix (intentionally scoped out, with reason).

#	Finding	Severity	Decision
1	Notification lead time too short for patients living far from clinic. The 3-hr lead time may not give enough time to configure a visit.	Medium	Will Fix & add a "notify me" toggle to the notification settings.
2	First-time staff need a trust cue and regional-language label on the Open OPD board; add Telugu and Hindi sub-labels.	High	Will Fix & add a "trust cue" icon to the Open OPD board.
3	Doctor-inactivity alert lacks elapsed time and indication of cause.	Low	Open — useful but not blocking; track for v1.2.
4	Next Patient button is vulnerable to inadvertent double-fire, risk of slipping a patient.	Critical	Will Fix & add a "confirm" button; server-side idempotency on the queue endpoint.
5	Average consultation time on the doctor console is low value to the doctor; is high on the (main) — board.	Low	Won't fix & add a "consultation time" metric to the v2 C
6	Dashboard UI is English-only. PHC staff comfort is critically constrained by the lack of Telugu and Hindi labels.	Critical	Will Fix & add Telugu and Hindi labels before any real-v

5. Observed Issues by Severity

Critical (blocks task completion or risks data integrity in production):

- Next Patient double-fire risk (Finding 4).
- English-only UI for non-English-speaking staff (Finding 6).

Major (causes hesitation, repeated mistakes):

- First-time staff trust cue on Open OPD (Finding 2).
- Notification lead-time configurability (Finding 1).

Minor (polish):

- Inactivity alert lacks elapsed time (Finding 3).
- Average consultation time placement on doctor view (Finding 5).

6. Scope of This Audit

The following were intentionally not assessed in this round and are scheduled for the pilot:

- WhatsApp Business API integration (V2 feature, out of V1 scope).
- Telugu and Hindi message rendering on the actual Android devices used in the field.
- Multi-day session behaviour across consecutive OPD days.
- Concurrent front-desk and doctor usage from a single tablet.
- Long-running stability over four-hour-plus sessions.

7. Conclusion

The system passes the developer self-test audit against the stated criterion of at least 80 percent of personas satisfied. Two critical findings (queue-advance idempotency and UI localization) must be addressed before any real-world deployment and are scheduled for v1.1. Four further findings are triaged: two will be fixed for v1.1, one is open for v1.2 consideration, and one is deferred to a future admin view. A five-user audit at the pilot site, Latha Children's Clinic, Attapur, is scheduled for pilot week one and will validate or supersede the findings recorded here.